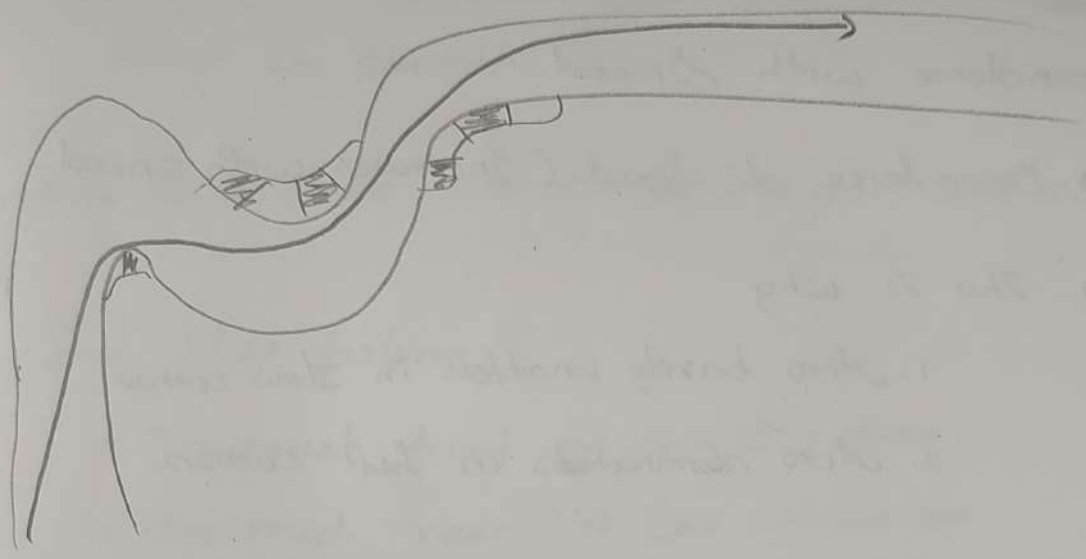
as early as possible. maximizing speed on the following straight.

- If gore a little wide on the first corner exit, you save little time but sacrifice in entry of next corner.



6. Aero Fundamentals.

1. Downforce

→ What is downforce?

↳ Air pushing the car to the ground at it forces the air to change direction.

→ Why it exists ?

↳ Air having momentum, ~~compresses~~ creates a pressure difference w.r.t the car creating a force.

↳ When the car does:

1. Deflect air upward (by rear ~~wor~~ wing).
2. Accelerate it under the floor
3. Compress it against body surfaces

it creates a pressure difference. ≐ force.

~~→ Dow~~

→ Dependence with Speed.

↳ Down force ∝ Speed. (Increases with speed).

↳ This is why

1. Aero barely matters in slow corners.
2. Aero dominates in fast corners.

↳ Downforce must be a parameter during the development of the model - as it help cornering and hence lap time. !!!

→ What more downforce does?

1. More vertical load on tyres.

↳ More load — more grip (but with diminishing results).

↳ For model.

Double load ≠ Double grip. !!

2. More potential grip.

3. More Stability.

→ Catch : Drag.

↳ Downforce brings Drag to the picture.

↳ Drag does the following:

1. Increases with speed
2. Reduces acceleration
3. Caps top speed.

↳ Increasing wing speed improves corner speed, but decreases speed in straight.

Downforce is a trade b/w corner time gain and straight time loss.

Downforce shifts balance:

1. Too much front → Car Oversteer.
2. Too much rear → Car Understeer.

2. Downforce scales with speed.

↳ Downforce roughly ~~goes~~ grows with square of speed.

→ What Happens?

↳ At high speeds:

1. Colide with more air ~~pressure~~ per second
2. Pressure difference grow dramatically.
3. ~~S~~ Floor tunnels suck more
4. Wings push harder.

→ Changes for corners:

↳ High Speed Corners:

- Massive downforce
- Aero determines maximum speed.
- If downforce is increased by 5%.
  1. Mid corner speed increases
  2. Lap time drops (reduces)

↳ Slow speed Corners:

- Less downforce contribution
- Grip is from tyres and setup.
- Speed is barely affected.

→ Force Mappings.

~~Trac~~ Track Segment	Dominant Factor.
Long Straights	Drag + Power
High Speed Corner	Downforce.
Medium Speed Corner	Mixed
Low Speed Corner	Mechanical Setup.
Traction Zone	Tyre + Load Transfer